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Education and Employment

Assistant Professor, Tsinghua University, 2025-*present*.

Associate Fellow, KICP, University of Chicago, 2021-2025.

PhD, Astrophysics, University of Michigan, Ann Arbor, 2017-2021.

MS, Astrophysics, University of Michigan, Ann Arbor, 2015-2017.

BS, Astronomy (Graduate with Honors), Peking University, 2011-2015

Scholarships & Awards

2021 - The IAU PhD Prize (Galaxies and Cosmology), IAU.

2021 - The ProQuest Distinguished Dissertation Awards, The University of Michigan.

2020 - Rackham Dissertation Fellowship, The University of Michigan.

2016-2019 - Rackham Travel Awards, The University of Michigan.

2015 - Weiming Physics Scholarship, School of Physics, Peking University.

2015 - Xingcheng Undergrad Research Forum, First Prize, School of Physics, Peking University.

2014 - Science Base Scholarship, Second Prize, Peking University.

2014 - NAOC Scholarship, National Astronomical Observatories, Chinese Academy of Sciences.

2013 - Lin Bridge Scholarship, Department of Astronomy, Peking University.

2013 - Zhang Wenxin Scholarship, School of Physics, Peking University.

Professional Services

2026-*present* MUST Extragalactic Science Working Group.

2025-*present* PhD Thesis Committee: Yaofei He (2025 THU); Yuqian Gui (2025 THU); Jingyi Zhang (2025 THU); Mengxuan Chen (2025 THU).

2025-*present* PhD Thesis Defense Committee: Hang Yang (2025 PMO); Dong Yang (2025 PKU); Yanling Chen (2025 THU).

2025-*present*, Solar Wind Magnetosphere Ionosphere Link Explorer (SMILE), Astrophysics Working Group, Member.

2024-*present*, Affordable Multiple Aperture Spectroscopy Explorer (AMASE), **CGM Working Group Co-Chair**.

2021-*present*, Hot Universe Baryon Survey (HUBS), Galactic Ecosystem Science Working Group, Member.

2023-*present*, External reviewer for HST.

2021-*present*, Referee for Science, SCPMA, AAS Journals (ApJ and ApJL), A&A, and MNRAS.

Research Advising

Postdoctoral

1. He Lin @ NJU->THU, 2026-*present*.

Graduate

8. Junle Wang @ BNU->THU, 2026-*present*.
7. Wenhao Wang @ XMU->THU, 2025-*present*.
6. Yuqian Gui @ THU, Co-advised with Dandan Xu, 2025-present.
5. Yuanzhen Han @ PMO, Co-advised with Shuinai Zhang and Li Ji, 2025-present.
4. Zeyang Pan @ UCAS, Co-advised Joel Bregman and Jifeng Liu 2021-*present*.
3. Suyash Kumar @ UChicago, Co-advised with Hsiao-Wen Chen, 2022-2025.
2. Cameron Pratt @ UMich, Co-advised with Joel N. Bregman, 2020-*2024*.
1. Hang Yang @ PMO, Co-advised with Joel Bregman and Li Ji, 2019-2025.

Undergraduate

5. Junle Wang @ BNU, 2025-2026.
4. Jinglun Dai @ THU, 2025-present.
3. Wenhao Wang @ XMU->THU, 2024-2025.
2. Eliana Schiller @ UChicago, Co-advised with Hsiao-Wen Chen, 2024-*present*.
1. Ryan Lindley @ UMich, 2019-2021.

Teaching

4. 2026 Spring, THU, instructor - Advanced Observational Astrophysics
3. 2025 Fall, THU, instructor (with Chris Ormel) - Special Topics in Theoretical Astrophysics
2. 2016 Fall, UMich, Graduate Student Instructor - Astro 102: Stars, Galaxies, and the Universe
1. 2016 Spring, UMich, Graduate Student Instructor - Astro 106: Aliens

Proposals and Grants

25. 2026 - *BTA 6-m Telescope*, 2027_1, BTA271.1303E, Main Program, 3 dark nights, **Co-I (PI: Wenhao Wang)**
"An Inclined View of Baryon Cycling in Dwarf Galaxies"

24. 2026 - *Lijiang 2.4-m Telescope*, 2026–2027, CHILI, 3 half nights, **PI**
"Baryon Cycling in Highly Inclined Dwarf Galaxies: Connecting the Interstellar and Circumgalactic Media"

23. 2026 - *Einstein Probe*, Cycle 3, Cycle3-0057, GO, 90 ks, **PI**
"Deciphering the Outflow and ISM/CGM Interplay toward the Galactic Center"

22. 2026 - *FAST*, 2026, SQB-2026-0172, 4 hrs, **PI**
"Resolving the ISM-CGM Connection in Highly-inclined Dwarfs"

21. 2025 - *FAST*, 2025, Key Program, 500 hrs, Co-I (PI: Jing Wang)
"FLOWER: FAST Long-exposure Observation of Wide-field HI Emission around/behind Resolved galaxies"

20. 2025 - *Einstein Probe*, Cycle 2, GO, 300 ks, Co-I (PI: Rui Huang)
"STP5/GES-WG: The Largest Hot Superbubbles Ever Seen in a Low Density Environment?"

19. 2025 - *Hubble Space Telescope*, Cycle 33, 18107, GO, 70 orbits, **Co-PI** (PI: Sean Johnson)
"Baryonic Escape, Abundances, and Thermodynamics in Cosmic Ecosystems (BEATriCE)"

18. 2025 - *Hubble Space Telescope*, Cycle 33, 18074, GO, 26 orbits, **Co-PI** (PI: Ava Polzin)
"PANCHO: Piloting a Novel Calibration for H2 Observations"

17. 2025 - *Hubble Space Telescope*, Cycle 33, 18039, GO, 9 orbits, Co-I (PI: Will Zhuoqi Liu)
"Confirming and Characterizing a Fast, Cool Inflow Toward a Luminous Quasar at $z = 0.78$"

16. 2024 - *Chandra*, Cycle 26, 26900274, GO, 190 ks, Co-I (PI: Joel Bregman)
"Identifying the Hosts of Hot Gas Absorption"

15. 2024 - *James Webb Space Telescope*, Cycle 3, 6665, GO, 16.2 hours, Co-I (PI: Alexander Beckett)
"DISCS: Direct Imaging Survey of Circumgalactic Structure"

14. 2024 - *Hubble Space Telescope*, Cycle 32, 17815, GO, 20 orbits, Co-I (PI: Alexander Beckett)
"DISCS: Direct Imaging Survey of Circumgalactic Structure"

13. 2024 - *Hubble Space Telescope*, Cycle 32, 17803, GO, 51 orbits, Co-I (PI: Sean Johnson)
"The first multi-phase survey of the circumgalactic medium of quasar hosts at $z < 1$"

12. 2024 - *FAST*, 2024, SQB-2024-0058, 15 hrs, Co-I (PI: Xuchen Lin) "Contiguous H I Mapping for Galaxies Evolving in 10^{13} $M_{\odot}$ Galaxy Groups"

11. 2023 - *Hubble Space Telescope*, Cycle 31, 17517, GO, 131 orbits, Co-I (PI: Hsiao-Wen Chen)
"CONTACT: Circumgalactic Observations of Nuv-shifted Transitions Across Cosmic Time"

10. 2022 - NASA Astrophysics Data Analysis Program, **Co-PI** (PI: Hsiao-Wen Chen), $454.6k
"Probing Feeding and Feedback in the Circumgalactic Medium Using OVI As a Tracer"

9. 2022 - *XMM-Newton*, AO22, 092249, GO, 133 ks, Co-I (PI: Joel Bregman)
"Closing the Baryon Census of Nearby Luminous Spirals with X-ray and tSZ Observations"

8. 2022 - *Hubble Space Telescope*, Cycle 30, 17116, GO, 30 orbits, Co-I (PI: Joel Bregman)
"The Hot Multi-Temperature Gaseous Halos of Galaxies and Groups"

7. 2022 - *Hubble Space Telescope*, Cycle 30, 17049, AR, **Co-PI** (PI: Ava Polzin)
   "Elucidating Galaxy Quenching with Absorption Probes of Halos around Low-mass Dwarfs"
6. 2021 - NASA Astrophysics Data Analysis Program, Co-I (PI: Joel Bregman), $692.8k ($80.0k goes to U. Chicago)
   "Resolving the Hot Halos to the Virial Radius and Beyond in the Local Universe"
5. 2019 - *Hubble Space Telescope*, Cycle 27, 15806, AR, **PI**, $97.6k
   "The Rotation, Accretion, and Mass of the Milky Way Warm Gas Disk and CGM"
4. 2018 - NASA Astrophysics Data Analysis Program, Co-I (PI: Joel Bregman), $526.4k
   "Hot Halos of Galaxies and Galaxy Groups-Reservoirs of Baryons and Metals"
3. 2018 - Magellan Telescope, Cycle 2018A, 2 nights, **PI**,
   "Host Galaxies of the Mg X/Ne VIII Absorption Systems: LBQS 1435-0134"
2. 2018 - Magellan Telescope, Cycle 2017B, 3 nights, **PI**,
   "Host Galaxies of the Mg X/Ne VIII Absorption Systems: PHL 1377"
1. 2017 - *Chandra* X-ray Observatory, Cycle 18, 50 ks, Co-I (PI: Joel Bregman),
   "Gaseous Galaxy Halos in the Transition Between Hot and Cold Mode Accretion."

Conferences & Talks

44. Oct. 2026 - Talk, "From Baryon Census to Gas Flows (at $z < 0.5$): the Dynamic Multiphase CGM", Dynamic Cosmic Structures (DCS), Hangzhou, China.
43. Aug. 2026 - Early Science Case, "The Circumgalactic Medium: Early Science Targets", AMASE, Kunming, China.
42. Jun. 2026 - Talk, "Tracing Multiphase Gas Cycling from Cosmic Noon to the Local Universe with CSST/MOS", CSST MOS, Shenzhen, China.
41. Jun. 2026 - Contributing talk, "Kinematic Signatures of Accretion and Feedback in the Circumgalactic Medium", Guoshoujing Conference, Haikou, China.
40. Jun. 2026 - Contributing talk, "Kinematic Diagnostics of the Circumgalactic Medium: Accretion and Feedback", HUBS SWG1, Guiyang, China.
39. Jun. 2026 - Talk, "Isolating the Milky Way's Hot CGM: Disentangling SWCX from Hot Gas Emission", NJU, Suzhou, China.
38. May 2026 - **Seminar**, "Multi-Scale Dynamic Gas in the Galactic Ecosystem", FDU, China.
37. Apr. 2026 - **Seminar**, "Multi-Scale Dynamic Gas in the Galactic Ecosystem", NAOC, China.
36. Dec. 2025 - **Invited talk**, "Mapping the Cool Circumgalactic Medium around Active Galaxy Nuclei", The 3rd AGN feedback conference, Nanjing, China.
35. Dec. 2025 - Talk, "The Circumgalactic Medium and Its Interface with the Interstellar Medium", AMASE-P Critical Design Review, Hongkong, China.
34. Nov. 2025 - Talk, "The Multiphase Circumgalactic Medium in UV Absorption", Diffuse Ionized Gas in Galaxies, Groups and Clusters – New Insights with New Space Observatories, Suzhou, China.
33. Nov. 2025 - Contributing talk, "Earth's Magnetospheric X-ray Emission Revealed in the X-LEAP program", SMILE Science Meeting, Chengdu, China.

32. Sept. 2025 - **Invited talk**, "FLOWER: FAST Long-exposure Observation of Wide-field HI Emission around/behind Resolved galaxies", FAST 2nd Science Conference, Pingtang, China.

31. Sept. 2025 - **Colloquium**, "Multi-phase Dynamic Gas in the Circumgalactic Medium", USTC, Hefei, China.

30. Aug. 2025 - Contributing talk, "The Multi-phase Diffuse Gas in the Milky Way", Russian-Chinese Meeting on High Energy Astrophysics, Moscow, Russia.

29. Jul. 2025 - Early Science Case, "Joint Analysis of Euclid Deep Field – the Formation of Large-scale Structure at the Cosmic Noon", CSST 2025 Science Meeting, Kunming, China.

28. Jun. 2025 - Contributing talk, "Probing Turbulence (?) in the CGM", Galactic Ecosystem in the Past Billions of Years, Beijing, China.

27. Jun. 2025 - **Invited talk**, "Multi-Wavelength View of the Milky Way Circumgalactic Medium", The 3rd HUBS workshop (2025), Beijing, China.

26. Jun. 2025 - **Invited talk**, "The Hot Diffuse Gas in the Milky Way and Solar Wind Charge Exchange", ISSI-BJ, Beijing, China.

25. May 2025 - Contributing talk, "Cool Gas in the Galactic Ecosystem", Guoshoujing Conference (Galaxy), Changsha, China.

24. May 2025 - AMASE Workshop, "Searching for Feedback or Accretion Features in the ISM-CGM Interface and CGM", XTU, Xiangtan, China.

23. Apr. 2025 - Colloquium, Dynamic Gas in the Galactic Ecosystem, Yunnan University, Kunming, China.

22. Apr. 2025 - Online talk, "X-LEAP: Decomposing the Milky Way Hot Gas Emission and the Foreground SWCX", SMILE Science Meeting, Leiden, Netherlands.

21. Apr. 2025 - Contributing talk, "Galactic Hot Gas and Solar Wind Charge Exchange: Insights from XMM-Newton", HUBS Galactic Science Work Group Meeting, Nanjing, China.

20. Apr. 2025 - Contributing talk, "(Potential?) Feedback Imprints in Diffuse Gas", Nearby Galaxy Symposium, THU, Beijing, China.

19. Mar. 2025 - Contributing talk, "X-LEAP: Decomposing the Milky Way Hot Gas Emission and the Foreground SWCX", Einstein Probe Diffuse Gas Workshop, Nanjing, China.

18. Nov. 2024 - Contributing talk, "The Multiphase Circumgalactic Medium from Low-mass Dwarfs to Massive Quiescent Galaxies at $z <\sim 1$", Resolving the Circumgalactic Medium and its Impact on Galaxy Evolution, Santa Cruz, Chile.

17. Aug. 2024 - Toward a Holistic Understanding of the Multi-scale, Multiphase Circumgalactic Medium, Aspen, Colorado, US.

16. Feb. 2024 - **Invited talk**, "On the Cosmic Ecosystem: Insights from the Multiphase Circumgalactic Medium", Diffuse Gas in Cosmic Ecosystems (DGCE) Science Interest Group, Online.

15. Apr. and Dec. 2023 - Seminars, FDU, XAO, XMU, SYSU, SHAO, SJTU, ZJU, THU, PKU, NAOC, PMO, USTC, NJU, China.

14. Dec. 2023 - Contributing talk, "The Mass Budget of Multiphase Circumgalactic Medium at $z \leq 1$", Resolving Galaxy Ecosystems Across All Scales, Hong Kong, China.

13. Nov. 2023 - **Invited talk**, Astro Galread Discussion Group, Princeton, US.
12. Feb. 2023 - Contributing talk, Oases in the Cosmic Desert, "On the thermodynamic properties of the cool CGM at z < 1", Tempe, Arizona, US.
11. Aug. 2022 - Instructor, The 1st "High-Resolution X-ray and UV (HiXUV) Spectroscopy Capability Building Workshops" , Virtual Workshop.
10. Jan. 2021 - **Invited talk**, OSU CCAPP Seminar, "Warm-Hot Circumgalactic Medium and its Co-Evolution with Galaxy Disk", Columbus, Ohio, US.
9. Nov. 2020 - Caltech CGM Seminar, "Warm-Hot Circumgalactic Medium and its Co-Evolution with Galaxy Disk", Pasadena, California, US.
8. Oct. 2019 - WIM Workshop, Contributing Talk, "The Warm Gas in the MW – Rotation, Accretion, and Mass", Green Bank, West Virginia, US.
7. Aug. 2018 - XXX IAU General Assembly, "The Mass and Column Densities of Galactic Gaseous Halos", Vienna, Austria.
6. Jul. 2018 - CGM Workshop 2018, Contributing Talk, "The Mass and Column Densities of Galactic Gaseous Halos", Northwestern University, Evanston, Illinois, US.
5. Mar. 2018 - Seminar, "Understanding the Galaxy Evolution with Gaseous Halos", Peking University, Beijing, China.
4. Mar. 2018 - Colloquium, "Understanding the Galaxy Evolution with Gaseous Halos", The Purple Mountain Observatory, Nanjing, China.
3. Jan. 2018 - 231st AAS Meeting, "The Mass and Absorption Columns of Galactic Gaseous Halos", Washington, DC, US.
2. Aug. 2017 - 16th HEAD Meeting, "Detection of Possible Intervening Mg X Associated with A Hot Gaseous Galaxy Halo", Sun Valley, Idaho, US.
1. Jul. 2014 - Fast Pulsar Symposium 3, "Analysis of Short Bursts in AXPs/SGRs", Shanghai, China.

Publications (First-Authored or Corresponding-Authored; *Mentored)

1. *Zeyang Pan, **Zhijie Qu**, Yuqi Gong, Tianran Sun, Joel N. Bregman, Yingjie Zhang, Li Ji, and Jifeng Liu. The XMM-Newton Line Emission Analysis Program (X-LEAP). III. Earth's Magnetospheric X-Ray Emission Revealed by 22 yr XMM-Newton Observations. *ApJ*, 1007(1):48, August 2026
2. **Zhijie Qu**, Hsiao-Wen Chen, Eliana Schiller, Jing Wang, and Max Gronke. Kinematics of H I and O VI Absorbers: Insights into the Turbulence Driver of the Multiphase Circumgalactic Medium. *ApJL*, 1000(1):L12, March 2026
3. *Hang Yang, **Zhijie Qu**, Joel N. Bregman, and Li Ji. Modelling the cool gas clumps in the circumgalactic medium. *MNRAS*, 538(3):1871–1883, April 2025
4. **Zhijie Qu**, Hsiao-Wen Chen, Sean D. Johnson, Gwen C. Rudie, Fakhri S. Zahedy, David DePalma, Joop Schaye, Erin T. Boettcher, Sebastiano Cantalupo, Mandy C. Chen, Claude-André Faucher-Giguère, Jennifer I. Hsiu Li, John S. Mulchaey, Patrick Petitjean, and Marc Rafelski. The Cosmic Ultraviolet Baryon Survey (CUBS). VII. On the Warm-hot Circumgalactic Medium Probed by O VI and Ne VIII at $0.4 < z < 0.7$. *ApJ*, 968(1):8, June 2024

5. **Zhijie Qu**, *Zeyang Pan, Joel N. Bregman, and Jifeng Liu. The XMM-Newton Line Emission Analysis Program (X-LEAP). II. The Multiscale Temperature Structures in the Milky Way Hot Gas. *ApJ*, 965(2):100, April 2024

6. *Zeyang Pan, **Zhijie Qu**, Joel N. Bregman, and Jifeng Liu. The XMM-Newton Line Emission Analysis Program (X-LEAP). I. Emission-line Survey of O VII, O VIII, and Fe L-shell Transitions. *ApJS*, 271(2):62, April 2024

7. **Zhijie Qu**, Hsiao-Wen Chen, Gwen C. Rudie, Sean D. Johnson, Fakhri S. Zahedy, David DePalma, Erin Boettcher, Sebastiano Cantalupo, Mandy C. Chen, Kathy L. Cooksey, Claude-André Faucher-Giguère, Jennifer I. Hsiu Li, Sebastian Lopez, Joop Schaye, and Robert A. Simcoe. The Cosmic Ultraviolet Baryon Survey (CUBS) - VI. Connecting physical properties of the cool circumgalactic medium to galaxies at $z \approx 1$. *MNRAS*, 524(1):512–528, September 2023

8. **Zhijie Qu**, Hsiao-Wen Chen, Gwen C. Rudie, Fakhri S. Zahedy, Sean D. Johnson, Erin Boettcher, Sebastiano Cantalupo, Mandy C. Chen, Kathy L. Cooksey, David DePalma, Claude-André Faucher-Giguère, Michael Rauch, Joop Schaye, and Robert A. Simcoe. The Cosmic Ultraviolet Baryon Survey (CUBS) V: on the thermodynamic properties of the cool circumgalactic medium at $z < 1$. *MNRAS*, 516(4):4882–4897, November 2022

9. **Zhijie Qu**, Dimitra Koutroumpa, Joel N. Bregman, Kip D. Kuntz, and Philip Kaaret. The Solar Cycle Temporal Variation of the Solar Wind Charge Exchange X-Ray Lines. *ApJ*, 930(1):21, May 2022

10. **Zhijie Qu** and Joel N. Bregman. Absorption Line Search through Three Local Group Dwarf Galaxy Halos. *ApJ*, 927(2):228, March 2022

11. **Zhijie Qu**, *Ryan Lindley, and Joel N. Bregman. The Warm Gas in the Milky Way: The Kinematical Model of C IV and Its Connection to Si IV. *ApJ*, 924(2):86, January 2022

12. **Zhijie Qu**, Rui Huang, Joel N. Bregman, and Jiang-Tao Li. An X-Ray- and SZ-bright Diffuse Source toward M31: A Local Hot Bridge. *ApJ*, 907(1):14, January 2021

13. **Zhijie Qu**, Joel N. Bregman, Edmund Hodges-Kluck, Jiang-Tao Li, and *Ryan Lindley. The Warm Gas in the MW: A Kinematical Model. *ApJ*, 894(2):142, May 2020

14. **Zhijie Qu** and Joel N. Bregman. The Warm Gaseous Disk and the Anisotropic Circumgalactic Medium of the Milky Way. *ApJ*, 880(2):89, Aug 2019

15. **Zhijie Qu**, Joel N. Bregman, and Edmund J. Hodges-Kluck. HST/COS Observations of the Warm Ionized Gaseous Halo of NGC 891. *ApJ*, 876(2):101, May 2019

16. **Zhijie Qu** and Joel N. Bregman. The Mass and Absorption Column Densities of Galactic Gaseous Halos. II. The High Ionization State Ions. *ApJ*, 862(1):23, July 2018

17. **Zhijie Qu** and Joel N. Bregman. The Mass and Absorption Columns of Galactic Gaseous Halos. *ApJ*, 856(1):5, March 2018

18. **Zhijie Qu** and Joel N. Bregman. A Hot Gaseous Galaxy Halo Candidate with Mg X Absorption. *ApJ*, 832:189, December 2016

19. **Zhijie Qu**, Zhaosheng Li, Yupeng Chen, Shi Dai, Long Ji, Renxin Xu, and Shu Zhang. The Short Bursts in SGR 1806-20, 1E 1048-5937, and SGR 0501+4516. *PASP*, 127(949):211, March 2015

Publications (Co-Authored)

20. Suyash Kumar, Hsiao-Wen Chen, Gwen C. Rudie, **Zhijie Qu**, Sean D. Johnson, Alexander Beckett, Sebastiano Cantalupo, Mandy C. Chen, David DePalma, Claude-André Faucher-Giguère, Sebastian Lopez, Nishant Mishra, Patrick Petitjean, Marc Rafelski, Michael Rauch, Joop Schaye, and Fakhri S. Zahedy. The Cosmic Ultraviolet Baryon Survey X: Density and Pressure Profiles of Clumpy Circumgalactic Gas at $z = 0.4$–0.7. *arXiv e-prints*, page arXiv:2610.36088, October 2026

21. Mandy C. Chen, Michael Rauch, **Zhijie Qu**, Gwen C. Rudie, Hsiao-Wen Chen, and James R. Beattie. From IFS Maps to 3D Structure I: Constraining Geometry in the Circumgalactic Medium. *arXiv e-prints*, page arXiv:2609.20929, September 2026

22. Ava Polzin, Hsiao-Wen Chen, **Zhijie Qu**, Erin Boettcher, Allison Strom, and Tsang Keung Chan. The Low-Mass Baryon Cycle in QUEST Dwarf Galaxies I: Sample Definition and First Results. *arXiv e-prints*, page arXiv:2608.15782, August 2026

23. Shengzhe Wang, Xin Wang, Hang Zhou, **Zhijie Qu**, Zhaozhou Li, Yuxuan Pang, Qianqiao Zhou, Shouyi Wang, Yangyao Chen, Yuguang Chen, Karl Glazebrook, Glenn G. Kacprzak, Nicha Leethochawalit, Houjun Mo, Themiya Nanayakkara, Huiyuan Wang, Weida Hu, Xunda Sun, Chao-Wei Tsai, and Hu Zhan. An emerging baryon cycle in a galaxy 500 million years after the Big Bang. *arXiv e-prints*, page arXiv:2608.09813, August 2026

24. Huanian Zhang, Lizhi Xie, **Zhijie Qu**, Dennis Zaritsky, Luis C. Ho, Min Bao, Fabio Fontanot, Michaela Hirschmann, Chenxu Liu, Gabriella De Lucia, Enci Wang, and Haoyan Zheng. Star formation powers optical line emission from the CGM. *Science Advances*, 12(38):eaec3939, September 2026

25. Andrew J. Fox, Jerry Kriss, Philipp Richter, J. Michael Shull, Frances Cashman, Sapna Mishra, Annelia Anderson, Nahum Arav, Ramona Augustin, Kathleen Barger, Michelle Berg, Rongmon Bordoloi, Sanchayeeta Borthakur, Joseph Burchett, Jane Charlton, Hsiao-Wen Chen, Christopher Churchill, Ryan Cooke, Annalisa de Cia, Gisella de Rosa, Romeel Davé, Yakov Faerman, Travis Fischer, David French, Farhan Hasan, Svea Hernandez, Cameron Hummels, Sean Johnson, Glenn Kacprzak, Vikram Khaire, Doyeon Avery Kim, Brad Koplitz, Varsha Kulkarni, Nicolas Lehner, Matilde Mingozzi, Talawanda Monroe, Sowgat Muzahid, Benjamin Oppenheimer, Molly Peeples, Céline Péroux, Patrick Petitjean, Andreea Petric, Max Pettini, **Zhijie Qu**, Kate Rowlands, Ravi Sankrit, Debopam Som, Raghunathan Srianand, Nicolas Tejos, Jason Tumlinson, Bart Wakker, and Jessica Werk. High-S/N Quasar Observations with HST/COS: Deep Fields for Spectroscopy. *arXiv e-prints*, page arXiv:2605.29949, May 2026

26. Zhuoqi Liu, Sean D. Johnson, Eric F. Bell, **Zhijie Qu**, Benoît Epinat, Hsiao-Wen Chen, Marc Rafelski, Jennifer I-Hsiu Li, Alexander Beckett, Mandy C. Chen, Sayak Dutta, David DePalma, Gwen C. Rudie, Joop Schaye, Patrick Petitjean, Sebastiano Cantalupo, Elise Fuller, Wolfram Kollatschny, Sebastián López, Nishant Mishra, Sowgat Muzahid, Andrea Travascio, and Fakhri S. Zahedy. A Comprehensive Study of Morphology and Kinematics in Extended Nebulae Around UV Luminous Quasars at $z \approx 1$. *ApJ*, 1008(2):247, September 2026

27. Yuxuan Zou, Feng Yuan, Suoqing Ji, Lin He, Zhiyuan Li, Yi Zhang, Johan Comparat, **Zhijie Qu**, and Taotao Fang. A Systematic Study of AGN Feedback in a Disk Galaxy Using MACER. II. Predictions of X-Ray Surface Brightness Profiles and Comparison with eROSITA Observations. *ApJ*, 1003(2):199, June 2026

28. Jiang-Tao Li, Renato A. Dupke, Yan Gong, **Zhijie Qu**, Weichen Wang, Xiaohu Yang, and Xiaodi Yu. Probing Large-scale Structure and the Multi-phase IGM at the Cosmic Noon—Insights from a Joint Survey with Euclid, CSST, JPCam, and JUST. *Research in Astronomy and Astrophysics*, 26(6):065021, June 2026

29. Jing Wang, Xuchen Lin, Ze-Zhong Liang, W. J. G. de Blok, Hong Guo, **Zhijie Qu**, Céline Péroux, Kentaro Nagamine, Luis C. Ho, Dong Yang, Simon Weng, Claudia del P. Lagos, Xinkai Chen, George Heald, J. Healy, Qifeng Huang, Peter Kamphuis, D. Kleiner, Di Li, Siqi Liu, F. M. Maccagni, Lister Staveley-Smith, Zherong Su, Freeke van de Voort, Fabian Walter, Fangxiong Zhong, and Siwei Zou. FEASTS and MHONGOOSE: H I Column Density Distribution at z = 0 for $N_{HI} > 10^{17.8}$ cm^{-2}. *ApJ*, 1001(1):123, April 2026

30. Yi Zhang, Soumya Shreeram, Gabriele Ponti, Johan Comparat, Andrea Merloni, **Zhijie Qu**, Jiangtao Li, Joel N. Bregman, and Taotao Fang. On the baryon budget in the X-ray-emitting circumgalactic medium of Milky Way-mass galaxies. *A&A*, 706:A102, February 2026

31. Sean D. Johnson, Nishant Mishra, Sowgat Muzahid, Gwen C. Rudie, Fakhri S. Zahedy, **Zhijie Qu**, Claude-André Faucher-Giguère, Jonathan Stern, Jennifer I.-Hsiu Li, Elise Fuller, Sebastiano Cantalupo, Hsiao-Wen Chen, Ahmad Kadri, Suyash Kumar, Zhuoqi (Will) Liu, and Gregory Walth. MUSEQuBES: Physical Conditions, Origins, and Multielement Abundances of the Circumgalactic Medium of an Isolated, Star-forming Dwarf Galaxy at z = 0.57. *ApJL*, 996(2):L30, January 2026

32. Rui Huang, Jiang-Tao Li, Wei Cui, **Zhijie Qu**, Joel N. Bregman, Xiang-Dong Li, Gabriele Ponti, and Q. Daniel Wang. An XMM—Newton view of the ANdromeda Galaxy as explored in a Legacy Survey (New-ANGELS) —II: luminosity function of X-ray sources. *MNRAS*, 544(4):3755–3773, December 2025

33. Huiyang Mao, Jiang-Tao Li, Xiaodi Yu, **Zhijie Qu**, Weizhe Liu, and Li Ji. Probing the He II Reionization ERa via Absorbing C IV Historical Yield (HIERACHY). IV. A Complex Redshifted Absorption System Intrinsic to Quasar. *ApJ*, 994(1):18, November 2025

34. David DePalma, Neeraj Gupta, Hsiao-Wen Chen, Robert A. Simcoe, Sergei Balashev, Erin Boettcher, Sebastiano Cantalupo, Mandy C. Chen, Françoise Combes, Claude-André Faucher-Giguère, Sean D. Johnson, Hans-Rainer Klöckner, Jens-Kristian Krogager, Jennifer I.-Hsiu Li, Sebastián López, Pasquier Noterdaeme, Patrick Petitjean, **Zhijie Qu**, Gwen C. Rudie, Joop Schaye, and Fakhri Zahedy. H I Properties of Field Galaxies at $z \approx 0.2$–0.6: Insights into Declining Cosmic Star Formation. *ApJL*, 993(1):L18, November 2025

35. Benjamin D. Oppenheimer, G. Mark Voit, Yannick M. Bahé, Nicolas Battaglia, Joel Bregman, Joseph N. Burchett, Dominique Eckert, Yakov Faerman, Justus Gibson, Cameron Hummels, Isabel Medlock, Daisuke Nagai, Mary Putman, **Zhijie Qu**, Ming Sun, Jessica K. Werk, and Yi Zhang. Introducing the Descriptive Parametric Model: gaseous profiles for galaxies, groups, and clusters. *MNRAS*, 543(3):2649–2669, November 2025

36. Cameron T. Pratt, **Zhijie Qu**, and Joel N. Bregman. Supervised Extraction of the Thermal Sunyaev–Zel'dovich Effect with a Three-dimensional Convolutional Neural Network. *ApJ*, 990(1):2, September 2025

37. Xiaodi Yu, **Zhijie Qu**, Zheng Cai, Jiang-Tao Li, Huiyang Mao, and Xin Wang. Probing the He II Reionization ERA via Absorbing C IV Historical Yield (HIERACHY). III. The C IV Absorber Catalog and Initial Results on Cosmic Abundance Evolution at $z \approx 3$–5. *ApJ*, 985(1):64, May 2025

38. Jiang-Tao Li, Xiaodi Yu, Huiyang Mao, Hanxiao Chen, Tiancheng Yang, **Zhijie Qu**, Fuyan Bian, Joel N. Bregman, Zheng Cai, Xiaohui Fan, Taotao Fang, Li Ji, Zhiyuan Ji, Sean D. Johnson, Guoliang Li, Weizhe Liu, Ying-Yi Song, Feige Wang, Tao Wang, Xin Wang, Christina Williams, Mingxuan Xu, Jinyi Yang, Yang Yang, and Xianzhong Zheng. Probing the He II re-Ionization ERa via Absorbing C IV Historical Yield (HIERACHY). II. Project Design, Current Status, and Examples of Initial Data Products. *ApJ*, 985(1):63, May 2025

39. Zhuoqi (Will) Liu, Sean D. Johnson, Jennifer I. Hsiu Li, Benoît Epinat, Gwen C. Rudie, Ana Monreal-Ibero, Sebastiano Cantalupo, **Zhijie Qu**, Mandy C. Chen, Wolfram Kollatschny, Sowgat Muzahid, Fakhri S. Zahedy, Elise Kesler, and Nishant Mishra. The Morphology and Kinematics of a Giant, Symmetric Nebula around a Radio-loud Quasar 3C 57: Extended Rotating Gas or Biconical Outflows? *ApJ*, 984(2):140, May 2025

40. Dong Yang, Jing Wang, **Zhijie Qu**, Zezhong Liang, Xuchen Lin, Simon Weng, Xinkai Chen, Barbara Catinella, Luca Cortese, D. B. Fisher, Luis C. Ho, Yingjie Jing, Fangzhou Jiang, Peng Jiang, Ziming Liu, Céline Péroux, Li Shao, Lister Staveley-Smith, Q. Daniel Wang, and Jie Wang. FEASTS Combined with Interferometry. IV. Mapping H I Emission to a Limit of $N_{HI} = 10^{17.7}$ cm^{-2} in Seven Edge-on Galaxies. *ApJ*, 984(1):15, May 2025

41. Xuchen Lin, Jing Wang, Lister Staveley-Smith, Suoqing Ji, Dong Yang, Xinkai Chen, Fabian Walter, Hsiao-Wen Chen, Luis C. Ho, Peng Jiang, Nir Mandelker, Se-Heon Oh, Bo Peng, Céline Péroux, **Zhijie Qu**, and Q. Daniel Wang. FEASTS Combined with Interferometry. III. The Low Column Density H I Around M51 and Possibility of Turbulent-mixing Gas Accretion. *ApJ*, 982(2):151, April 2025

42. Jing Wang, Dong Yang, Xuchen Lin, Qifeng Huang, **Zhijie Qu**, Hsiao-wen Chen, Hong Guo, Luis C. Ho, Peng Jiang, Zezhong Liang, Céline Péroux, Lister Staveley-Smith, and Simon Weng. FEASTS: Radial Distribution of H I Surface Densities Down to 0.01 $M_\odot$ pc^{-2} of 35 Nearby Galaxies. *ApJ*, 980(1):25, February 2025

43. Nishant Mishra, Sean D. Johnson, Gwen C. Rudie, Hsiao-Wen Chen, Joop Schaye, **Zhijie Qu**, Fakhri S. Zahedy, Erin T. Boettcher, Sebastiano Cantalupo, Mandy C. Chen, Claude-André Faucher-Giguére, Jenny E. Greene, Jennifer I. Hsiu Li, Zhuoqi (Will) Liu, Sebastian Lopez, and Patrick Petitjean. The Cosmic Ultraviolet Baryon Survey (CUBS). IX. The Enriched Circumgalactic and Intergalactic Medium Around Star-forming Field Dwarf Galaxies Traced by O VI Absorption. *ApJ*, 976(1):149, November 2024

44. Suyash Kumar, Hsiao-Wen Chen, **Zhijie Qu**, Mandy C. Chen, Fakhri S. Zahedy, Sean D. Johnson, Sowgat Muzahid, and Sebastiano Cantalupo. On the Nature of the C IV-bearing Circumgalactic Medium at $z \sim 1$. *The Open Journal of Astrophysics*, 7:94, October 2024

45. Jing Wang, Xuchen Lin, Lister Staveley-Smith, Dong Yang, Fabian Walter, Zezhong Liang, Yong Shi, Jian Fu, Hong Guo, Luis C. Ho, Shu-ichiro Inutsuka, Fangzhou Jiang, Peng Jiang, **Zhijie Qu**, and Li Shao. FEASTS Combined with Interferometry. II. Significantly Changed H I Surface Densities and Even More Inefficient Star Formation in Galaxy Outer Disks. *ApJ*, 973(1):15, September 2024

46. Joel N. Bregman, Oleg Y. Gnedin, Patrick O. Seitzer, and **Zhijie Qu**. A Type Ia Supernova near a Globular Cluster in the Early-type Galaxy NGC 5353. *ApJL*, 968(1):L6, June 2024

47. Jiang-Tao Li, Li-Yuan Lu, **Zhijie Qu**, Robert A. Benjamin, Joel N. Bregman, Ralf-Jürgen Dettmar, Jayanne English, Taotao Fang, Judith A. Irwin, Yan Jiang, Hui Li, Guilin Liu, Paul Martini, Richard J. Rand, Yelena Stein, Andrew W. Strong, Carlos J. Vargas, Q. Daniel Wang, Jing Wang, Theresa Wiegert, Jianghui Xu, and Yang Yang. eDIG-CHANGES. II. Project Design and Initial Results on NGC 3556. *ApJ*, 967(2):78, June 2024

48. Sean D. Johnson, Zhuoqi (Will) Liu, Jennifer I. Hsiu Li, Joop Schaye, Jenny E. Greene, Sebastiano Cantalupo, Gwen C. Rudie, **Zhijie Qu**, Hsiao-Wen Chen, Marc Rafelski, Sowgat Muzahid, Mandy C. Chen, Thierry Contini, Wolfram Kollatschny, Nishant Mishra, Patrick Petitjean, Michael Rauch, and Fakhri S. Zahedy. Discovery of Optically Emitting Circumgalactic Nebulae around the Majority of UV-luminous Quasars at Intermediate Redshift. *ApJ*, 966(2):218, May 2024

49. Jennifer I. Hsiu Li, Sean D. Johnson, Erin Boettcher, Sebastiano Cantalupo, Hsiao-Wen Chen, Mandy C. Chen, David R. DePalma, Zhuoqi (Will) Liu, Nishant Mishra, Patrick Petitjean, **Zhijie Qu**, Gwen C. Rudie, Joop Schaye, and Fakhri S. Zahedy. The Cosmic Ultraviolet Baryon Survey (CUBS). VIII. Group Environment of the Most Luminous Quasars at z $\approx$ 1. *ApJ*, 965(2):143, April 2024

50. *Cameron T. Pratt, **Zhijie Qu**, Joel N. Bregman, and Christopher J. Miller. Optimizing NILC Extractions of the Thermal Sunyaev–Zel'Dovich Effect with Deep Learning. *ApJ*, 964(2):122, April 2024

51. Mandy C. Chen, Hsiao-Wen Chen, Michael Rauch, **Zhijie Qu**, Sean D. Johnson, Joop Schaye, Gwen C. Rudie, Jennifer I. Hsiu Li, Zhuoqi (Will) Liu, Fakhri S. Zahedy, Sebastiano Cantalupo, and Erin Boettcher. An Ensemble Study of Turbulence in Extended QSO Nebulae at z $\approx$ 0.5–1. *ApJ*, 962(1):98, February 2024

52. Hsiao-Wen Chen, **Zhijie Qu**, Michael Rauch, Mandy C. Chen, Fakhri S. Zahedy, Sean D. Johnson, Joop Schaye, Gwen C. Rudie, Erin Boettcher, Sebastiano Cantalupo, Claude-André Faucher-Giguère, Jenny E. Greene, Sebastian Lopez, and Robert A. Simcoe. The Cosmic Ultraviolet Baryon Survey: Empirical Characterization of Turbulence in the Cool Circumgalactic Medium. *ApJL*, 955(1):L25, September 2023

53. Joel Bregman, Renyue Cen, Yang Chen, Wei Cui, Taotao Fang, Fulai Guo, Edmund Hodges-Kluck, Rui Huang, Luis C. Ho, Li Ji, Suoqing Ji, Xi Kang, Xiaoyu Lai, Hui Li, Jiangtao Li, Miao Li, Xiangdong Li, Yuan Li, Zhaosheng Li, Guiyun Liang, Helei Liu, Wenhao Liu, Fangjun Lu, Junjie Mao, Gabriele Ponti, **Zhijie Qu**, Chenxi Shan, Lijing Shao, Fangzheng Shi, Xinwen Shu, Lei Sun, Mouyuan Sun, Hao Tong, Junfeng Wang, Junxian Wang, Q. Daniel Wang, Song Wang, Tinggui Wang, Weiyang Wang, Zhongxiang Wang, Dandan Xu, Haiguang Xu, Heng Xu, Renxin Xu, Xiaojie Xu, Yongquan Xue, Hang Yang, Feng Yuan, Shuinai Zhang, Yuning Zhang, Zhongli Zhang, Yuanyuan Zhao, Enping Zhou, and Ping Zhou. Scientific objectives of the Hot Universe Baryon Surveyor (HUBS) mission. *Science China Physics, Mechanics, and Astronomy*, 66(9):299513, September 2023

54. Rui Huang, Jiang-Tao Li, Wei Cui, Joel N. Bregman, Xiang-Dong Li, Gabriele Ponti, **Zhijie Qu**, Q. Daniel Wang, and Yi Zhang. An XMM-Newton View of the Andromeda Galaxy as Explored in a Legacy Survey (New-ANGELS). I. The X-Ray Source Catalog. *ApJS*, 268(1):36, September 2023

55. Li-Yuan Lu, Jiang-Tao Li, Carlos J. Vargas, Rainer Beck, Joel N. Bregman, Ralf-Jürgen Dettmar, Jayanne English, Taotao Fang, George H. Heald, Hui Li, **Zhijie Qu**, Richard J. Rand, Michael Stein, Q. Daniel Wang, Jing Wang, Theresa Wiegert, and Yun Zheng. eDIG-CHANGES I: extended Hα emission from the extraplanar diffuse ionized gas (eDIG) around CHANG-ES galaxies. *MNRAS*, 519(4):6098–6110, March 2023

56. Mandy C. Chen, Hsiao-Wen Chen, Michael Rauch, **Zhijie Qu**, Sean D. Johnson, Jennifer I. Hsiu Li, Joop Schaye, Gwen C. Rudie, Fakhri S. Zahedy, Erin Boettcher, Kathy L. Cooksey, and Sebastiano Cantalupo. Empirical constraints on the turbulence in QSO host nebulae from velocity structure function measurements. *MNRAS*, 518(2):2354–2372, January 2023

57. Joel N. Bregman, Edmund Hodges-Kluck, **Zhijie Qu**, Cameron Pratt, Jiang-Tao Li, and Yansong Yun. Hot Extended Galaxy Halos around Local L* Galaxies from Sunyaev-Zeldovich Measurements. *ApJ*, 928(1):14, March 2022

58. *Cameron T. Pratt, **Zhijie Qu**, and Joel N. Bregman. The Resolved Sunyaev-Zel'dovich Profiles of Nearby Galaxy Groups. *ApJ*, 920(2):104, October 2021

59. Xiaodi Yu, Jiang-Tao Li, **Zhijie Qu**, Ian U. Roederer, Joel N. Bregman, Xiaohui Fan, Taotao Fang, Sean D. Johnson, Feige Wang, and Jinyi Yang. Probing the He II re-Ionization ERa via Absorbing C IV Historical Yield (HIERACHY) I: A strong outflow from a z $\sim$ 4.7 quasar. *MNRAS*, 505(3):4444–4455, August 2021

60. Joel N. Bregman, Michael E. Anderson, Matthew J. Miller, Edmund Hodges-Kluck, Xinyu Dai, Jiang-Tao Li, Yunyang Li, and **Zhijie Qu**. The Extended Distribution of Baryons around Galaxies. *ApJ*, 862(1):3, July 2018

61. Zhaosheng Li, **Zhijie Qu**, Li Chen, Yanjun Guo, Jinlu Qu, and Renxin Xu. An Ultra-low-mass and Small-radius Compact Object in 4U 1746-37? *ApJ*, 798(1):56, January 2015

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